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SECUREMAILSCOPE | CRYPTOGRAPHIC SECURITY POSTURE ASSESSMENT REPORT
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Capture File: mail-lab-1790669433159-0b228432-820e-47bf-8264-5ee98ebe11c4.pcapng
Session ID: 1790669447491-49b49c26-2253-40e7-869c-c8b1a59c80ef
Analysis Time: 2026-09-29T08:10:47.492Z

------------------------------ 1. EXECUTIVE SUMMARY ----------------------------------
Security Score: 100 / 100 (Grade: A, Posture: Excellent)
Policy Compliance: Compliant / Fully Consistent
Capture Integrity: Verified AES-256-GCM Configuration Manifest
Executive Summary: AI assessment unavailable. Deterministic rule-based cryptographic evalua

------------------------------ 2. CAPTURE OVERVIEW ----------------------------------
Packet Count: 18 packets
Capture Duration: 1.0 seconds
Capture Format: PCAPNG
Network Flow: 172.30.0.2:50000 -> 172.30.0.3:587

---------------------- 3. EMAIL PROTOCOL & TCP STREAM ANALYSIS -------------------------
Detected Protocol: SMTP on Port 587 (high confidence)
STARTTLS State: SECURE UPGRADE
STARTTLS Description: STARTTLS command executed and successfully upgraded plaintext channel to
STARTTLS Advertised: YES | Negotiated: YES | Required: YES
TCP Streams Count: 18 stream(s) reconstructed (complete)

------------------------------ 4. TLS LAYER ANALYSIS ----------------------------------
Negotiated TLS Version: TLS1.3
Selected Cipher Suite: TLS_AES_256_GCM_SHA384
Key Exchange Algorithm: ECDHE (Named Group: X25519)
Forward Secrecy (PFS): ENABLED (Ephemeral Key Exchange)
Handshake Completion: YES (Handshake Completed Successfully)

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## 5. X.509 CERTIFICATE ANALYSIS

Subject Common Name: mail.smtp.lab.local
Issuing CA: SecureMailScope Test CA (CA-ISSUED)
Validity Window: 2025-09-29 to 2027-09-29
Expiration Check: VALID
Public Key: RSA 2048 bits
Signature Algorithm: SHA256withRSA
Trust Chain Status: TRUSTED / VALID CHAIN

## 6. CRYPTOGRAPHIC POSTURE ACROSS 17 SECURITY CATEGORIES

[01] Protocol Security : [PASS ] Session successfully established cryptog
[02] TLS Version : [PASS ] Modern TLS protocol version TLS1.3 obser
[03] Cipher Suite : [PASS ] Strong modern AEAD cipher suite TLS_AES_
[04] Cryptographic Strength : [PASS ] Cryptographic parameters meet NIST SP 80
[05] Key Exchange : [PASS ] Ephemeral key exchange observed, protect
[06] Forward Secrecy : [PASS ] Perfect Forward Secrecy is guaranteed by
[07] Certificate Security : [PASS ] Certificate is signed by a recognized Ce
[08] Certificate Chain : [PASS ] Complete verifiable certificate chain ob
[09] Certificate Expiration : [PASS ] Certificate is within its active validit
[10] Public Key Algorithm : [PASS ] Algorithm RSA is approved under NIST SP
[11] Public Key Length : [PASS ] RSA 2048-bit meets current NIST 112-bit
[12] Signature Algorithm : [PASS ] Cryptographically secure signature algor
[13] STARTTLS Security : [PASS ] STARTTLS capability advertised and TLS u
[14] Configuration Compliance : [PASS ] All observable cryptographic settings sa
[15] Deprecated Cryptography : [PASS ] No deprecated or retired cryptographic p
[16] TLS Anomalies : [PASS ] Clean TLS handshake and execution profil
[17] Configuration vs Observed mismatch : [PASS ] Configured parameters verified identical

## 7. CRYPTOGRAPHIC SECURITY STRENGTH (NIST SP 800-57 Part 1 Rev. 5)

* Public Key : RSA | Strength: 112 bits (Acceptable through 2030) | [A
* Bulk Encryption : AES-256 | Strength: 256 bits | [STRONG]
* Key Exchange Group : ECDHE | Strength: 128 bits | [STRONG]
* Signature Hash : SHA256withRSA | Strength: 128 bits | [STRONG]

---------- 8. CONFIGURATION COMPLIANCE (CONFIGURED POLICY VS OBSERVED) ----------

| | | | |
|---|---|---|---|
| Protocol | : Configured: SMTP | Observed: SMTP | [MATCH] |
| Port | : Configured: 587 | Observed: 587 | [MATCH] |
| STARTTLS | : Configured: Enabled | Observed: Advertised | [MATCH] |
| TLS Version | : Configured: TLS1.3 | Observed: TLS1.3 | [MATCH] |
| Cipher Suite | : Configured: TLS_AES_256_GCM_SHA384 | Observed: TLS_AES_256_GCM_SHA384 | [ |
| Key Exchange | : Configured: ECDHE | Observed: ECDHE | [MATCH] |
| Named Group | : Configured: X25519 | Observed: X25519 | [MATCH] |
| Forward Secrecy | : Configured: Enforced | Observed: Enforced | [MATCH] |
| Certificate Present | : Configured: Yes | Observed: Yes | [PASS] |
| Certificate Validity | : Configured: Valid | Observed: Valid | [PASS] |

------------------- 9. SECURITY SCORE DEDUCTIONS BREAKDOWN ------------------------------

Overall Score: 100/100 | Grade: A | Posture: Excellent

  * No score deductions. All observable parameters satisfy standards.

-------------------------- 10. SECURITY FINDINGS (0) --------------------------

  * Zero security findings identified in this session.

## 11. ANOMALY DETECTION

* No anomalous downgrade attempts or protocol discrepancies identified.

## 12. AI SECURITY ASSESSMENT

AI assessment unavailable. Showing deterministic security assessment.

## 13. PRIORITIZED REMEDIATION RECOMMENDATIONS

## 14. STANDARDS & NORMATIVE REFERENCES

* NIST SP 800-52 Rev. 2 : Guidelines for the Selection, Configurat (Section 3.1 - 3.4)
* NIST SP 800-57 Part 1 Rev. 5 : Recommendation for Key Management: Crypt (Section 5.6.1 Tabl
* RFC 8996 : Deprecating TLS 1.0 and TLS 1.1 (BCP 195 (Sections 1 - 2)
* RFC 8446 : The Transport Layer Security (TLS) Proto (Section 4.4 & Appendi
* RFC 3207 : SMTP Service Extension for Secure SMTP o (Sections 4 & 5)
* CISA Guidance : CISA Capacity Building Guidance for Secu (Section 4.2)
* CA/Browser Forum : Baseline Requirements for the Issuance a (Section 6.1.5 & 6.3.2